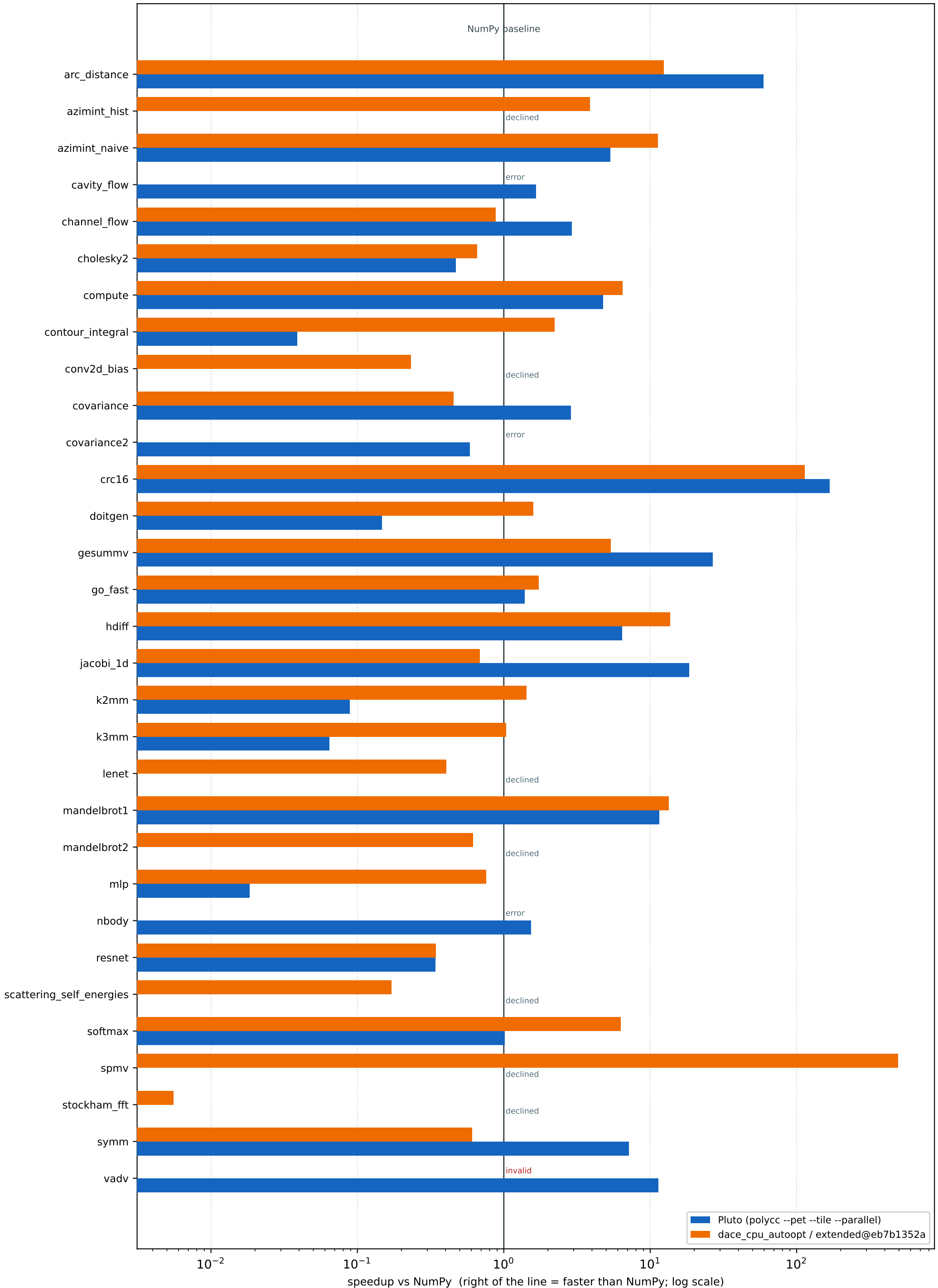


NPBench PolyBench-derived kernels
 preset paper, REPEAT=50 - VERIFICATION RUN, not a performance measurement
 DaCe column = dace_cpu_autoopt / extended@eb7b1352a



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DaCe column = dace_cpu_autoopt / extended@eb7b1352a
↓ = faster than NumPy, ↑ = slower; only validated results carry a speedup

kernel	NumPy (baseline)	Pluto	DaCe CPU	DaCe variant	notes
arc_distance	0.46 s	↓59.2	↓12.3	auto_opt	
azimint_hist	14.82 ms	declined	↓3.9	auto_opt	pluto: polycc failed on azimint_hist_pluto_reference.c: [Clan] Error: variab
azimint_naive	0.49 s	↓5.3	↓11.2	auto_opt	
cavity_flow	1.61 s	↓1.7	error		dace: no result row and no recognised failure in the log
channel_flow	2.86 s	↓2.9	↑1.1	auto_opt	
cholesky2	21.81 ms	↑2.1	↑1.5	auto_opt	
compute	0.44 s	↓4.8	↓6.4	auto_opt	
contour_integral	0.49 s	↑25.7	↓2.2	auto_opt	
conv2d_bias	7.59 s	declined	↑4.3	auto_opt	pluto: clang did not finish within 900s on conv2d_pluto.c
covariance	28.03 ms	↓2.9	↑2.2	auto_opt	
covariance2	5.53 ms	↑1.7	error		dace: ModuleNotFoundError: No module named 'npbench.benchmarks.polybench.co
crc16	1.41 s	↓167	↓113	auto_opt	
doitgen	0.24 s	↑6.8	↓1.6	auto_opt	
gesummv	0.20 s	↓26.8	↓5.4	auto_opt	
go_fast	0.11 s	↓1.4	↓1.7	auto_opt	
hdiff	0.15 s	↓6.4	↓13.7	auto_opt	
jacobi_1d	0.26 s	↓18.4	↑1.5	auto_opt	
k2mm	0.17 s	↑11.3	↓1.4	auto_opt	
k3mm	0.18 s	↑15.5	↓1.0	auto_opt	
lenet	1.46 s	declined	↑2.5	auto_opt	pluto: polycc did not finish within 900s on lenet_pluto_reference.c
mandelbrot1	0.88 s	↓11.6	↓13.3	auto_opt	
mandelbrot2	0.37 s	declined	↑1.6	auto_opt	pluto: polycc failed on mandelbrot2_pluto_reference.c: [Clan] Error: non-aff
mlp	23.66 ms	↑54.6	↑1.3	auto_opt	
nbody	0.30 s	↓1.5	error		dace: encountered in File "/users/levwidmer/npbench/npbench/benchmarks/nbo
resnet	1.54 s	↑2.9	↑2.9	auto_opt	
scattering_self_energies	1.06 s	declined	↑5.8	auto_opt	pluto: polycc failed on scattering_self_energies_pluto_reference.c: [Clan] E
softmax	0.83 s	↓1.0	↓6.3	auto_opt	
spmv	0.24 s	declined	↓494	auto_opt	pluto: polycc failed on spmv_pluto_reference.c: [Clan] Error: syntax error a
stockham_fft	95.48 ms	declined	↑180	auto_opt	pluto: polycc failed on stockham_fft_pluto_reference.c: [Clan] Error: variab
symm	7.27 s	↓7.1	↑1.6	auto_opt	
vadv	0.54 s	↓11.4	invalid		dace: ran but did not validate against the NumPy reference